Your grade: 90%

Your latest: **90%** • Your highest: **90%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Suppose that $\lim_{x\to a} f(x) = 32$, $\lim_{x\to a} g(x) = 68$, and $\lim_{x\to a} h(x) = 10$. 1 / 1 point

Then $\lim_{x\to a} \dfrac{f(x)+g(x)}{h(x)}$ is equal to which of the following?

- (•) 10
- ( ) The limit is undefined.
- ( ) The limit cannot be determined from the given function.
- ( ) $\dfrac{f(a)+g(a)}{h(a)}$

Nice work

Since the limits of all three functions exist and the limit of $h(x)$ as $x \to a$ is nonzero, we can apply the sum and quotient laws to compute the limit.

2. Gary is simplifying the expression for a function $f(x)$. What is wrong with his work? 1 / 1 point

$$f(x) = \frac{x^2-5x+6}{3x-6}$$
$$f(x) = \frac{(x-2)(x-3)}{3x-6}$$
$$= \frac{(x-2)(x-3)}{3(x-2)}$$
$$= \frac{x-3}{3}.$$

Therefore, Gary concludes, $f(x) = \dfrac{x-3}{3}$.

- ( ) There is nothing wrong with his work.
- (•) Gary has to note that x cannot equal 2 using his simplified expression.
- ( ) He factored incorrectly.
- ( ) The factors of $(x-2)$ cannot be canceled from the fraction.

Nice work

When canceling the factors of $(x-2)$ from the fraction, Gary has to note that x cannot equal 2, since you are not allowed to divide by 0. Therefore, he should write $f(x) = \frac{x-3}{3}$ and $x \neq 2$ for his simplified expression for the function.

3. $f(x) = (2x^2 - 4x - 8)^6$ 1 / 1 point

Find $\lim_{x\to 3} f(x)$.

64

Nice work

Since $f(x)$ is a polynomial, we can use the direct substitution property. The limit is

$$\lim_{x\to 3} f(x) = f(3) = (2(3)^2 - 4(3) - 8)^6 = (-2)^6 = 64$$

4. Evaluate the limit, if it exists. 1 / 1 point

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

$$\lim_{u\to -2} \sqrt{u^4+3u+6}$$

4

Nice work

Apply the Root Law

$$\lim_{u\to -2} \sqrt{u^4+3u+6} = \sqrt{\lim_{u\to -2} u^4+3u+6}$$

Since the expression inside the radical is a polynomial, we can use the direct substitution property to find that the limit is equal to

$$\sqrt{(-2)^4+3(-2)+6} = \sqrt{16} = 4$$

5. Evaluate the limit, if it exists. 1 / 1 point

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

$$\lim_{x\to -6} \frac{2x+12}{|x+6|}$$

DNE

Nice work

By definition of the absolute value function,

$\dfrac{2x+12}{|x+6|} = \dfrac{2x+12}{x+6}$ when $x+6 \geq 0$, i.e. when $x \geq -6$.

$\dfrac{2x+12}{|x+6|} = \dfrac{2x+12}{-(x+6)}$ when $x+6 < 0$, i.e. when $x < -6$.

We can cancel out a common factor ($x+6$) from the numerator and denominator,

$\dfrac{2x+12}{|x+6|} = 2$ when $x \geq -6$ and $\dfrac{2x+12}{|x+6|} = -2$ when $x < -6$.

Thus $\lim_{x\to -6^-} f(x) = -2 \neq 2 = \lim_{x\to -6^+}$, so $\lim_{x\to -6} f(x)$ does not exist.

6. True or False? 1 / 1 point

If $-x^2+4x-3 \leq f(x) \leq x^2-3x+4$

then $\lim_{x\to 1} f(x) = 0$

- ( ) True
- (•) False

Nice work

$\lim_{x\to 1} -x^2+4x-3 = 0$

$\lim_{x\to 1} x^2-3x+4 = 2$

Therefore, we cannot use to squeeze theorem to determine $\lim_{x\to 1} f(x)$.

7. If $2x^2 - 14 \leq f(x) \leq 9x - 18$, what is $\lim_{x\to 4} f(x)$? 1 / 1 point

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

18

Nice work

$\lim_{x\to 4} 2x^2 - 14 = 18$

$\lim_{x\to 4} 9x - 18 = 18$

Therefore, by the squeeze theorem, $\lim_{x\to 4} f(x) = 18$

8. Using the Squeeze Theorem, what is $\lim_{x\to 0^+} |x| \cos\left(\frac{1}{\sqrt{x}}\right)$? 0 / 1 point

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

DNE

Try again

$-1 \leq \cos\left(\frac{1}{\sqrt{x}}\right) \leq 1$

$-|x| \leq |x| \cos\left(\frac{1}{\sqrt{x}}\right) \leq |x|$

$\lim_{x\to 0^+} -|x| = \lim_{x\to 0^+} |x| = 0$

$\lim_{x\to 0^+} |x| \cos\left(\frac{1}{\sqrt{x}}\right) = 0$

9. Determine the limit: 1 / 1 point

$$\lim_{x\to 3} \frac{\sin(x-3)}{\sqrt{x+6}}$$

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

0

Nice work

The function is a rational function, and $x = 3$ is in its domain. Therefore, we can use the direct substitution property to find the limit.

$$\lim_{x\to 3} \frac{\sin(x-3)}{\sqrt{x+6}} = \frac{\sin(3-3)}{\sqrt{3+6}} = \frac{\sin(0)}{3} = 0$$

10. Evaluate the limit, if it exists. 1 / 1 point

If your answer is not a whole number, enter it as a decimal. If the limit does not exist, enter "DNE"

$$\lim_{x\to -4} \frac{\sqrt{x^2+9}-5}{x+4}$$

-0.8

-0.8

Nice work

We can't apply the Quotient Law, since the limit of the denominator is 0.

First, we must rationalize the numerator.

$$\lim_{x\to -4} \frac{\sqrt{x^2+9}-5}{x+4} = \lim_{x\to -4} \frac{\sqrt{x^2+9}-5}{x+4} \cdot \frac{\sqrt{x^2+9}+5}{\sqrt{x^2+9}+5}$$
$$= \lim_{x\to -4} \frac{x^2+9-5^2}{(x+4)(\sqrt{x^2+9}+5)} = \lim_{x\to -4} \frac{x^2-16}{(x+4)(\sqrt{x^2+9}+5)}$$

Now factor the numerator and cancel out common factors from the denominator.

$$= \lim_{x\to -4} \frac{(x+4)(x-4)}{(x+4)(\sqrt{x^2+9}+5)} = \lim_{x\to -4} \frac{x-4}{\sqrt{x^2+9}+5}$$

Now we can use the Quotient, Power, and Root Laws to compute the limit.

$$= \frac{\lim_{x\to -4} x-4}{\lim_{x\to -4} \sqrt{x^2+9}+5} = \frac{-8}{\sqrt{\lim_{x\to -4} x^2+9}+5}$$
$$= \frac{-8}{\sqrt{(-4)^2+9}+5} = \frac{-8}{10} = -0.8$$